Watcharapon Thodraksa

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SKILLS

Programming Languages: Python, SQL, TypeScript, JavaScript, Java

ML / Computer Vision: PyTorch, ONNX, OpenVINO, DINOv2, EfficientNet, YOLO, OpenCV, LightGBM, Sentinel-1 SAR, Sentinel-2 NDVI, rasterio

MLOps/DevOps: Kubernetes (k3s), Helm, Docker, MLflow, Flux GitOps, GitHub Actions, GitLab CI, FastAPI, Grafana, Prometheus, Kaggle GPU

LLM / Backend: LangGraph, LangChain, pgvector, BGE-M3, Ollama, RAG, NestJS, Spring Boot, Kafka, PostgreSQL

EDUCATION

Kasetsart University, Chalermphrakiat Sakon Nakhon Campus
Bachelor of Science (B.Sc.) | Major: Computer Science (Co-op Program) | Cumulative GPA: 3.10/4.00 | Graduated 2025

PROFESSIONAL EXPERIENCE

AI/ML Engineer (Contract) **TRR** 2026 – Present
Computer vision and satellite ML for a sugar mill group in Thailand.

- **CaneSight (CCTV dump-yard monitoring):** built the system that turns every truck dump into a result card for the mill's app: dust level 0–3, leafy-cane check, burnt or fresh, video clip and the matched truck. Seven top cameras and one side camera, served through a REST API on a CPU-only k3s server inside the mill (OpenVINO, ONNX).
- **MLOps for CaneSight:** MLflow model registry on k3s, training on Kaggle with pass/fail gates written before each test, promotion by registry alias with human approval, automatic deploy with checksum check and rollback, Flux GitOps with GitHub Actions checks, daily score-drift monitoring and Telegram alerts.
- **Burnt vs fresh cane classifier:** EfficientNet-B0 trained on labels from 25,980 weighbridge tickets; AUC 0.994 on 483 trucks from days the model never saw.
- **Leafy-cane detection:** fine-tuned DINOv2; correct verdict on 23 of 23 held-out truck frames at 1.8 s per image on CPU. Rejected a newer candidate that failed its gate on 300 new trucks.
- **CropScan (harvest monitoring from satellite):** Sentinel-2 with Sentinel-1 radar fallback under cloud; F1 0.883 on 8,924 held-out fields (FastAPI, rasterio, Railway).
- **Yield Pro (field health and yield):** SAR-to-NDVI gap-filling model with 95.7% of values within ±0.10 on 129k held-out pixels; yield ranges per field (LightGBM, PyTorch, MLflow).
- **CaneGate (weighbridge truck inspection):** cane grading and Thai licence-plate OCR (0 wrong reads on 60 hard images), shipped as Docker images and used at the mill.

DevOps & Backend Engineer (Intern) **Sycapt** Jan 2026 – May 2026

- Built an on-premises RAG knowledge system for a network with no internet access: LangGraph agents, BGE-M3 embeddings, pgvector hybrid search with re-ranking, Ollama; ingests documents, Mantis tickets and email (n8n); deployed on Kubernetes.
- Ran four LLMs on a single 16 GB GPU by swapping models through 120 GB of RAM.
- Ran the Kubernetes, Docker and Helm deployments; designed the GitLab CI/CD pipeline (build, test, Docker build, staging, production with automatic rollback) and set up Grafana and Prometheus monitoring.

Programmer (Intern) **Sycapt** Nov 2025 – Jan 2026

- Developed backend services for a payment service provider handling 1M+ transactions per day (PromptPay, Alipay).
- Built an event-driven Kafka connector microservice (Java 21, Spring Boot, ELK) and a map dashboard for 15,423 payment terminals nationwide (Leaflet, FastAPI).

PROJECTS

Portfolio website: portfolio-watcharapon.vercel.app (case studies for the work above)

YouTube Shorts Automation:

Python · YouTube Analytics API · ffmpeg

- Pipeline that reads channel analytics, picks topics, generates and uploads clips on schedule; 3.34M views across 189 clips.

Teaching and Awards:

- Invited lecturer at Kasetsart University: workshops on React Native with the Gemini API and on Git/GitHub.
- 1st Runner-up, KUSE AI Hackathon 2025 · Winner, UI Design-athon 2025 · Participant, RERU Cyber Hackathon 2025.